RMO(INDIA) – 1985(1)

Max Mark – 100 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Find the last two digits i.e. the digits in the unit's place and ten's place of the number $7^{100} - 3^{100}$.
2. If a >b > 0, show that $\dfrac{1+b+b^2+...+b^9}{1+b+b^2+...+b^{10}} > \dfrac{1+a+a^2+...+a^9}{1+a+a^2+...+a^{10}}$.
3. Express $(\sqrt{2}-1)^{10}$ in the form of $\sqrt{k}-\sqrt{k-1}$ where k is a positive integer. (The square roots need not be irrational)
4. For what values of x where $-\pi < x \le \pi$ the following inequality is satisfied. $\left|\sin x - \cos x\right| \le \dfrac{1}{\sqrt{2}}$.
5. Show that sum of the lengths of the medians of a triangle is smaller than the perimeter of the triangle but is greater than three fourths of the perimeter of the triangle.
6. A and B are two distinct points neither of which is on the line L and both are on the same side of L. A point P moves on the line L. Determine the position of P so that $l(PA)+l(PB)$ is minimum.
7. Show that all the rectangles (i) having the same perimeter the square has the greatest area, (ii) having the same area the square has the least perimeter.
8. Show that there exist positive irrational numbers a and b (not necessarily distinct) such that a^b is a rational number. (hint: Make use of the number $\sqrt{2}^{\sqrt{2}}$.
9. If a and b are acute angles with a < b show that

   (i) $a - \sin a < b - \sin b$ (ii) $\sqrt{k}\tan a - a < \tan b - b$
10. Show that $(a+b+c)^9 - a^9 - b^9 - c^9$ is divisible by $(a+b+c)^3 - a^3 - b^3 - c^3$.
11. From the first hundred positive integers 51 integers are arbitrarily chosen. Show that this set of 51 integers always contain a pair of numbers such that one of them is divisible by other.
12. If a and b are the roots of the equation $x^2 - 6x + 1 = 0$, show that for any positive integer, $a^n + b^n$ is an integer which is not divisible by 5.

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